# Fetch My Route!

User Interface Design Document (UIDD)

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# 1. Introduction

## 1.1. Purpose of This Document

1.1.1. The purpose of this document is to outline the user-interface (UI) design for the SDS Accessibility GPS website. It will describe both visual and interactive standards for the UI, and ultimately showcase how the development team achieved these requirements. The intended audience of this document is the product owner, stakeholder (who is also the customer), and the UI developer to ensure all parties are in agreement of how the final application will appear on both desktop and mobile web browsers.

## 1.2. References

1.2.1. SRS Document - UMBC Accessibility GPS

1.2.2. UMBC Accessibility Routes Map

# 2. User Interface Standards

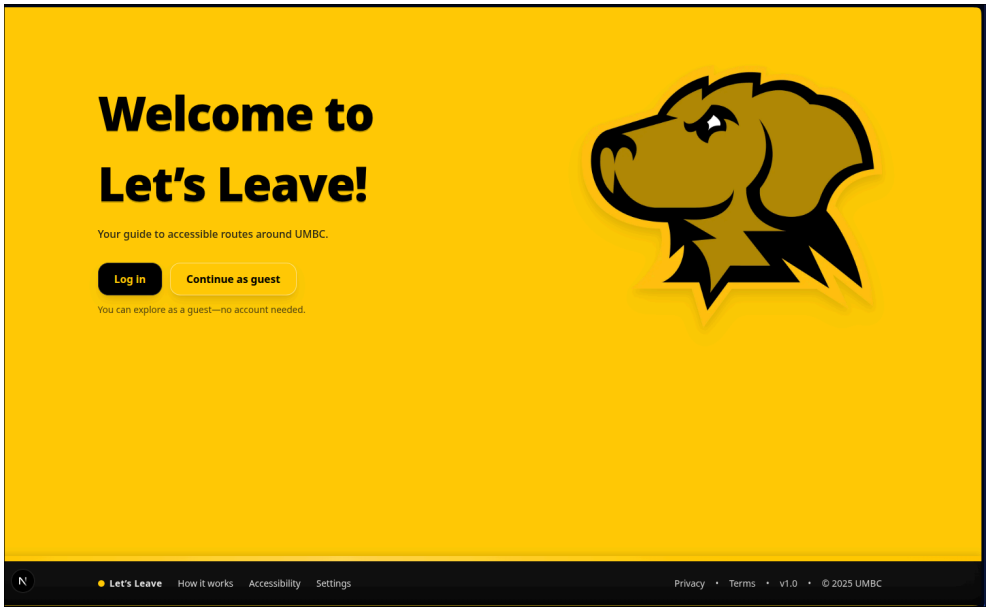
This section describes the UI standards and specifications for this project. While many other groups have the same project, this implementation is unique and better than every other group. (The previous statement has not been tested, evaluated, or fact-checked in any way. Source: Trust me )

## 2.1 Welcome Page

The first thing the user sees when entering our website is the Welcome Page. This page is vibrant, in an attempt to draw the eye of the user, yet simple with buttons and interactables so as to not overwhelm the user. From the welcome page the user may:

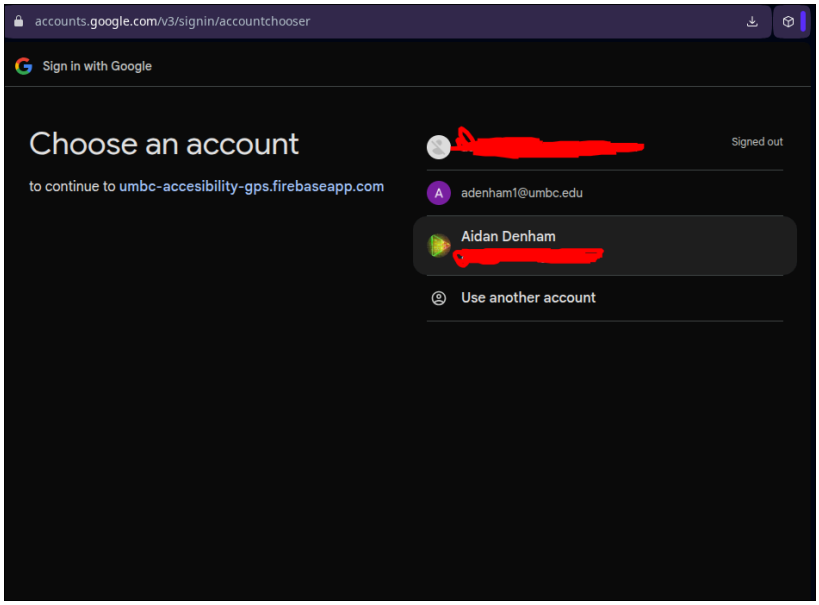
1. Continue to the Main Page as a guest
2. Login with their google email account (UMBC domains only) and continue to the Main Page
3. View any of the footer buttons described in section 2.1.3

### 2.1.1 Welcome Splash Screen




### 2.1.2 Login Google OAuth Page

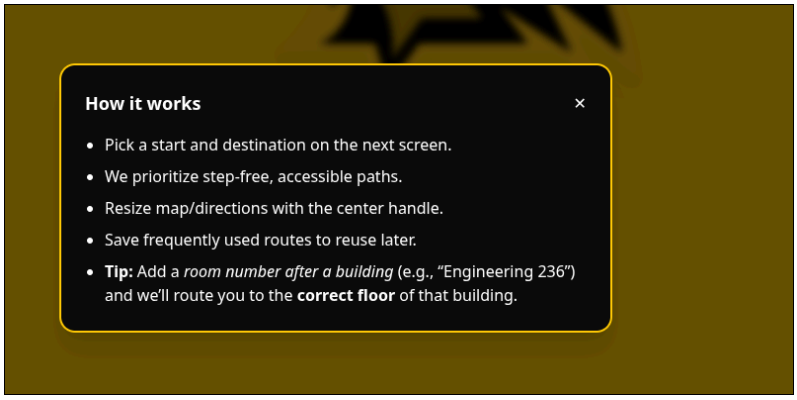
Our website features login via the Google OAuthentication API. While this allows for theoretically any valid gmail to be used, our backend only accepts emails with a @umbc.edu domain. This way, Google's API success validates the email is real, and our backend validates that the domain is correct, culminating in the resulting login being an already-existing valid UMBC email.



### 2.1.3 Footer Buttons

This page is also host to many footer buttons, which provide extra detailing or utility to enhance the user experience. None of the footer buttons affect the core functionality of the website, hence their presence in a footer, but allow for optional “extra” features. While the footer buttons are visible on all main screens (2.1, 2.2), this is the first occurrence of them, hence the stylistic choice of mentioning them here.

#### 2.1.3.1 How it Works

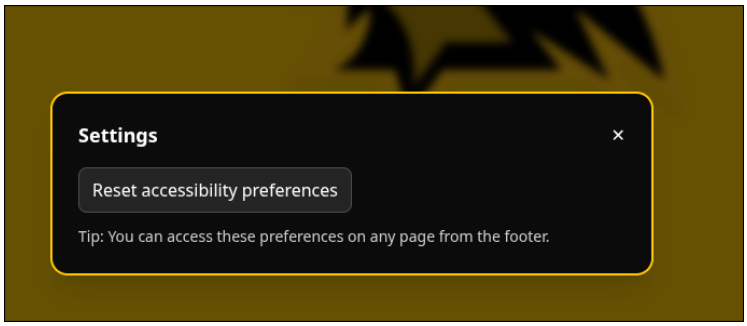



This page describes, in short form, how to use and operate the website. Further instructions can be found in the README page of our project GitHub, but these points can guide the user on the basics of the site if they get stuck.

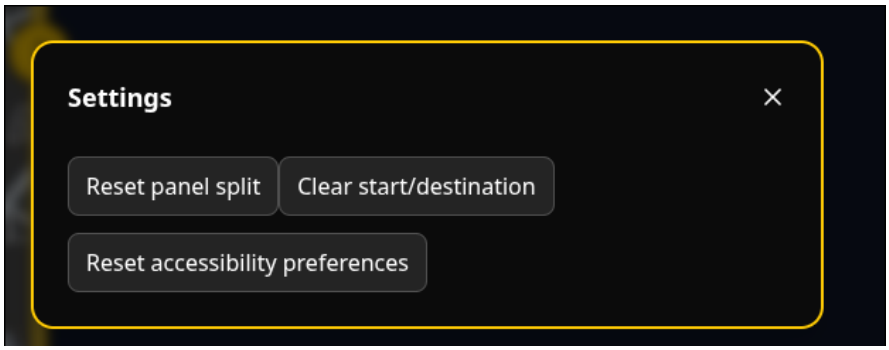
#### 2.1.3.2 - Accessibility

This page describes basic accessibility options for the site. As the project is targeted towards users with mobility impairments, a working accessibility page is vital to ease of use. This button provides a centralized place for users to change settings as they see fit.

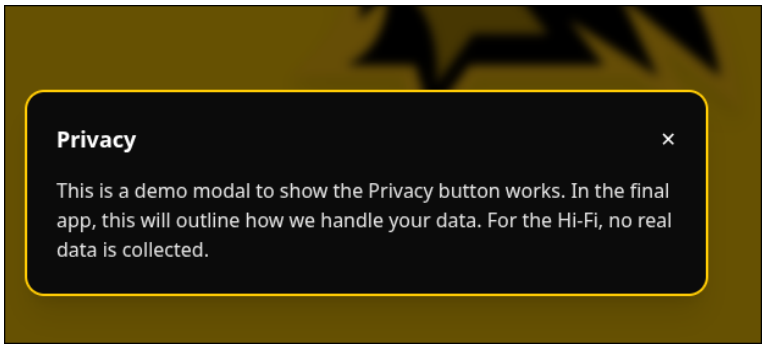
#### 2.1.3.3 - Settings




As of now, there are not many settings for the user to test. However, there will be more in the final version of this project, and therefore they deserve their own page. One such example of a setting is to reset the accessibility preferences that the user set in the previous button. **Note:** While the “Settings” button appears the same across screens, the contents of the button change depending on what screen the user is on. For example, on the Main Page (2.2) where routing is involved, there are extra buttons for route settings. See below:

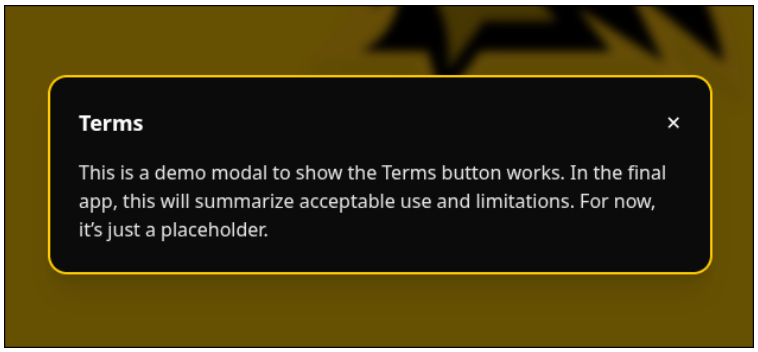



#### 2.1.3.4 - Privacy




This screen will display the project’s privacy policy, which has not been fleshed out yet.

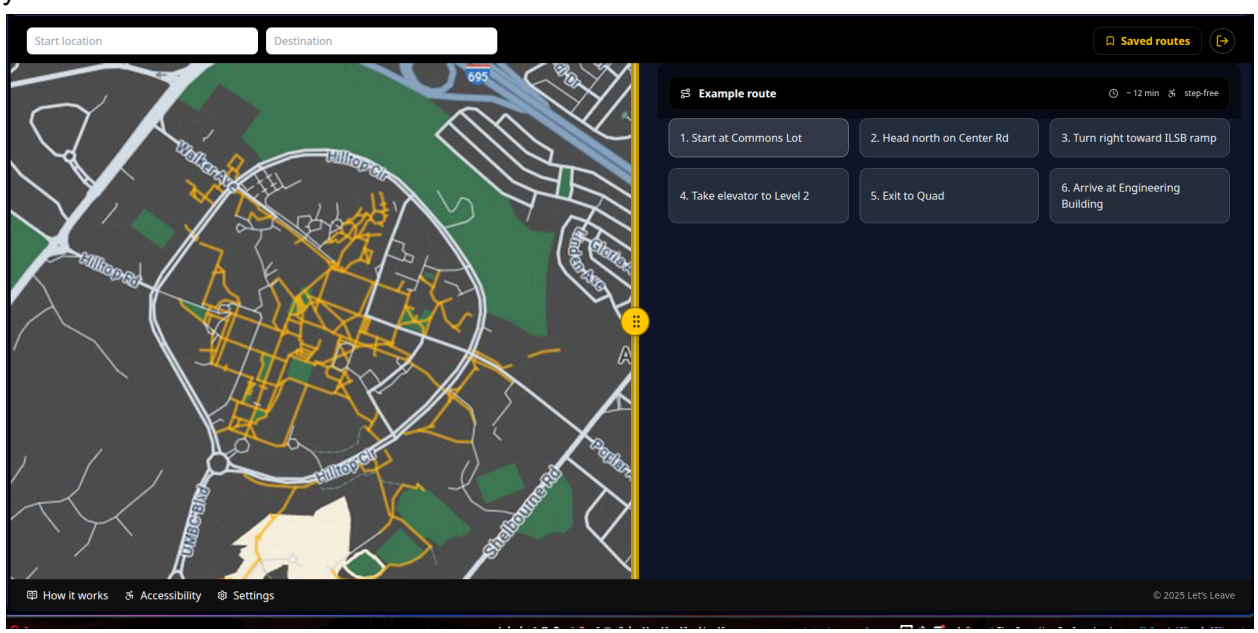
#### 2.1.3.5 - Terms & Conditions




This screen will display the project's terms and conditions, which would be the responsibility of the stakeholder to populate before the website gets deployed.

## 2.2 Main Page

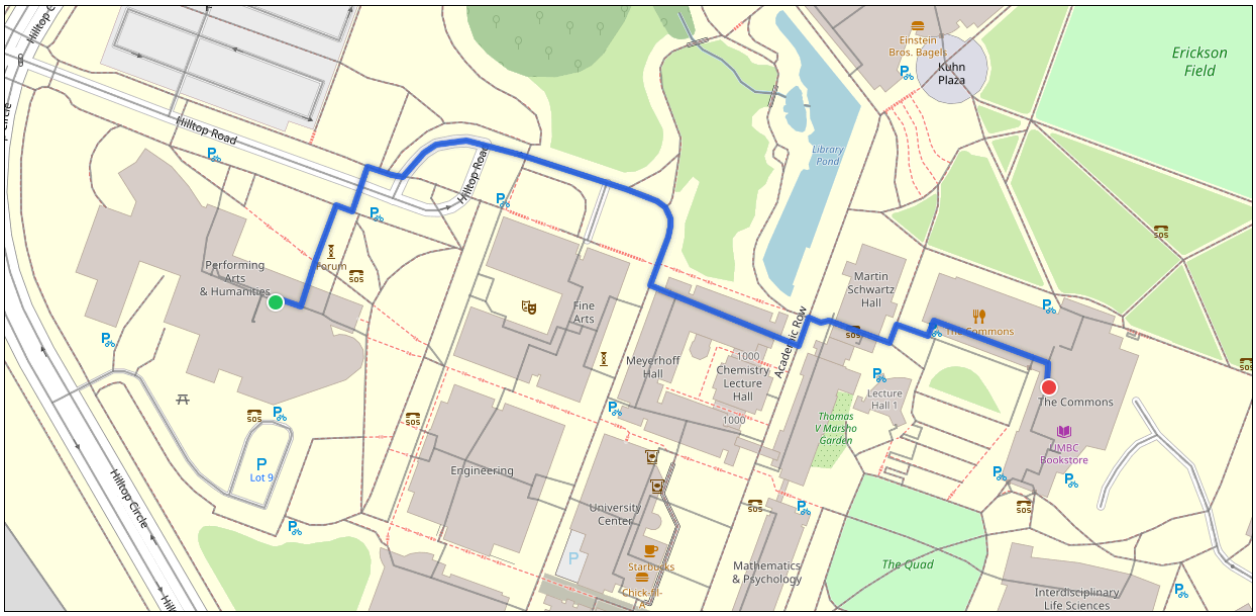
The main page of this website features a map (temporary map shown in illustration, final integration with working map not yet finalized at this point), the appropriate buttons for interacting with the map, and a split panel with written instructions on how to proceed across the map. Each of these parts will be explained more thoroughly in their own sections, but when combined these two screens can be resized, split, and joined as the user wishes through the yellow button in the center. The footer buttons from Section 2.1 are also visible here.



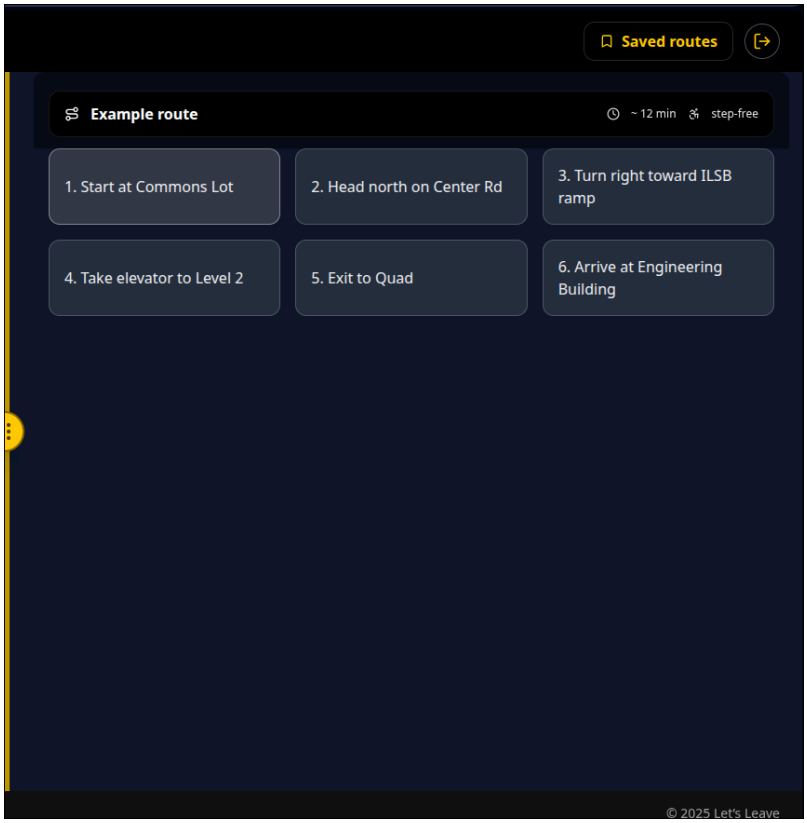

### 2.2.1 - Map Side

The Map Side of the main page hosts the OpenStreepMap Leaflet (referenced from the SDS), and on the map is a visual walkthrough of the user's selected route. There also exist text fields for the user to input their start and end locations on campus, which will visually alter the map and draw a route, if one exists. As the map appearing above is only a still image (frontend

integration with the backend is an ongoing effort), I will include an image of the actual map and what it looks like with the user entering a route. The green dot represents the starting spot, while the red dot represents the end location. Therefore, the below route is directing the user from the Performing Arts & Humanities Building to The Commons, perhaps for a midnight snack after our 7pm class ends.

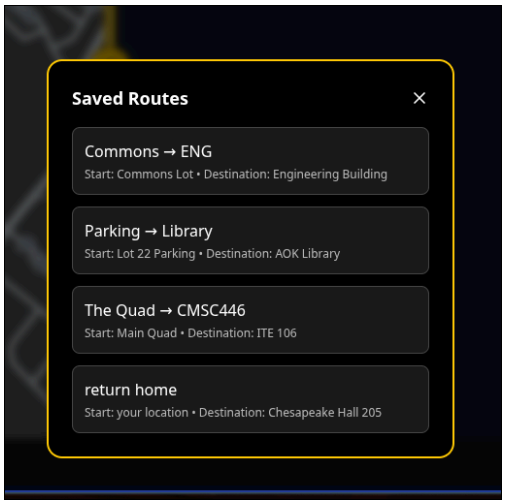


### 2.2.2 - Split Panel Side




This side of the screen features written instructions on how to navigate the user, if the user prefers reading this as opposed to looking at the map. This is just an example route, but these route instructions will generate dynamically based on the user's inputted route on the Map Side (See 2.2.1)

#### 2.2.2.1 - Saved Routes Button

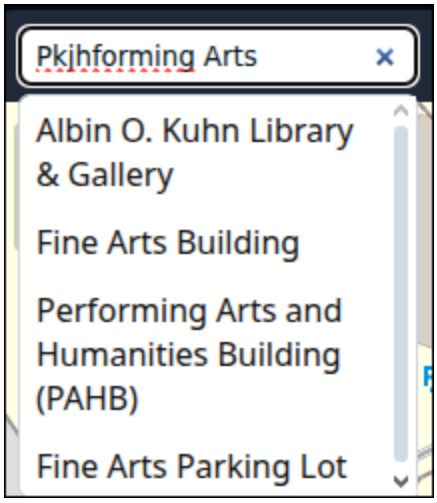


Clicking the Saved Routes button (seen in 2.2.2) opens up a panel with a list of sample routes which a user can select. This functionality is only available to users who have logged in, as guest accounts do not have their data saved to the database.

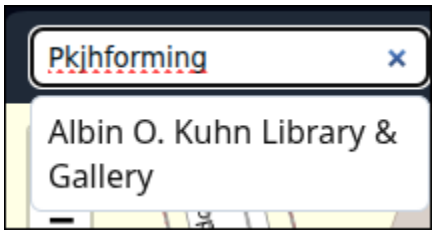
## 2.3 Error Handling

There are few places in the project where users physically enter data, which minimizes the opportunities for user error. However, error can still occur in two places: login email and start/end building search. Login email errors occur when a user attempts to login using the Google OAuth page and type in incorrect email addresses. Since this service is not managed by the project, error handling is also implemented by Google. Therefore the only real source of error in this project is when a user attempts to enter a building that is either out of scope of this project (Not a UMBC building) or misspells a word.

The former error is a security risk to the project and likely an intentional choice. This is mitigated by both the building-search and building-click functionality are limited to within the bounds of the UMBC campus, along with the scrollable bounds for the map. This ensures that users cannot even access data outside of UMBC, so routing attempts will be unsuccessful. For the latter case, where a user misspells a word, the project has software bounds to mitigate this. The user input is routed to a custom search function, which uses a combination of tokenization, substring matching, and the Levenshtein Distance of each substring, in order to match partially correct inputs to search suggestions of campus buildings. With these algorithms implemented, it is much harder to incorrectly misspell a campus building. For example, incorrectly spelling the Performing Arts and Humanities Building renders the below suggestions



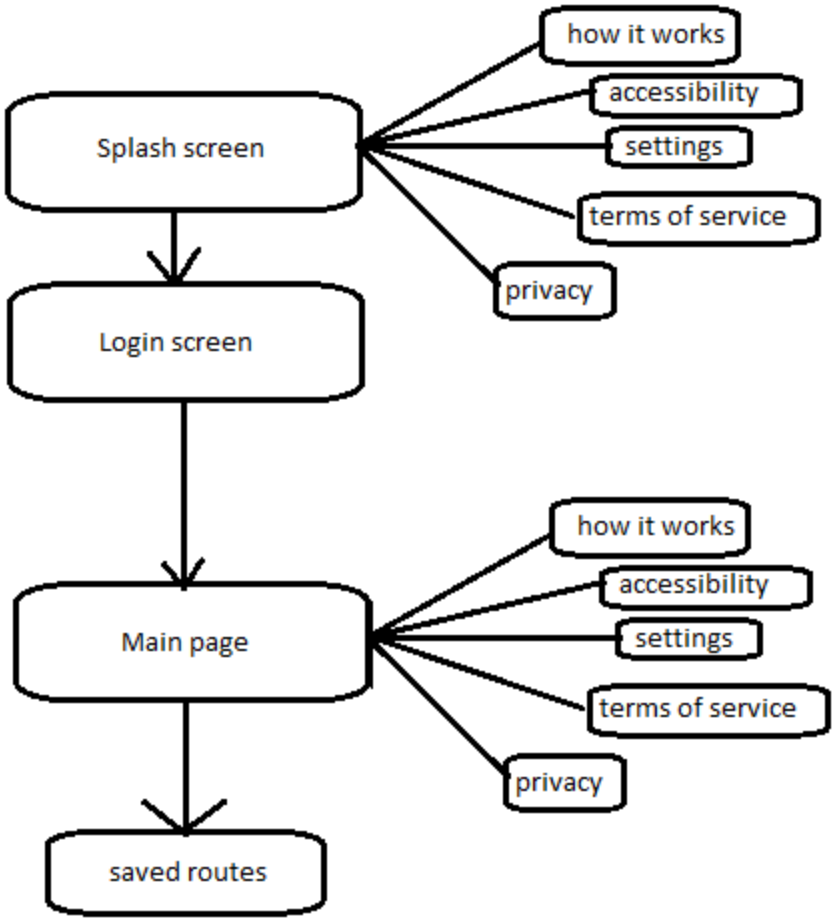

, however removing the phrase “Arts” renders the incorrect suggestion below.




In the case where there are no suggestions available, the user gets a popup which states so. This is the catch-all case where even the search suggestions cannot determine the correct building. In the event this case happens, the user can divert to the “Map Click” functionality where they can click the building to route instead of continuing to attempt to spell the building name.

# 3. User Interface Walkthrough

3.1: The following diagram summarizes how users move between all major screens in the system:

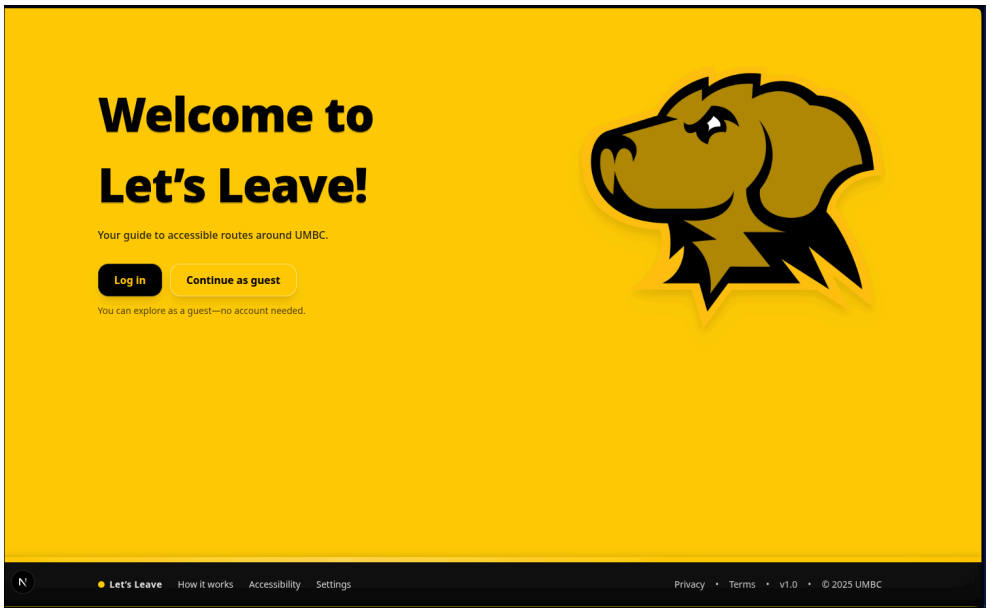



**Description:**
This diagram represents the complete UI flow for the application. Users start at the Splash Screen, log in or continue as a guest, then arrive at the Main Page. The footer buttons (How It Works, Accessibility, Privacy, Settings, Terms & Conditions) are accessible from any screen except the Login page.

## 3.2 Screen Walkthrough

Every major screen in the system is shown below with a description of its layout, purpose, and interactive components. Each figure number matches the navigation diagram.

## Figure 1. Splash Screen

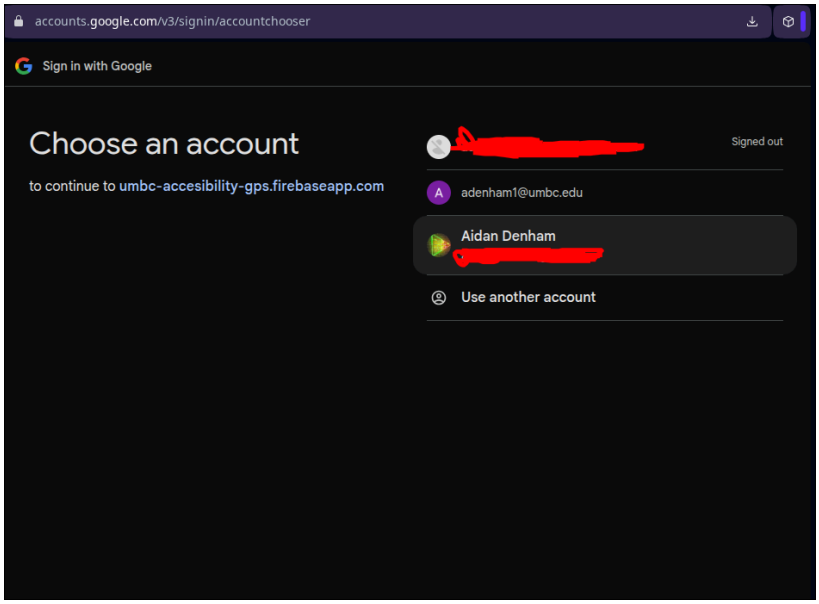



**Description:**
The Splash Screen is the first page users see. It displays the UMBC logo and title, centered on the screen. At the bottom is a navigation bar with the standardized footer buttons ("How It Works," "Accessibility," "Settings," "Privacy," "Terms & Conditions"). Two main actions are displayed:

- **Login with Google** — sends the user to the OAuth login screen.
- **Continue as Guest** — bypasses login and brings the user directly to the Main Page.

All elements are centered and sized for comfortable interaction on mobile and desktop.

## Figure 2. Login (Google OAuth) Screen

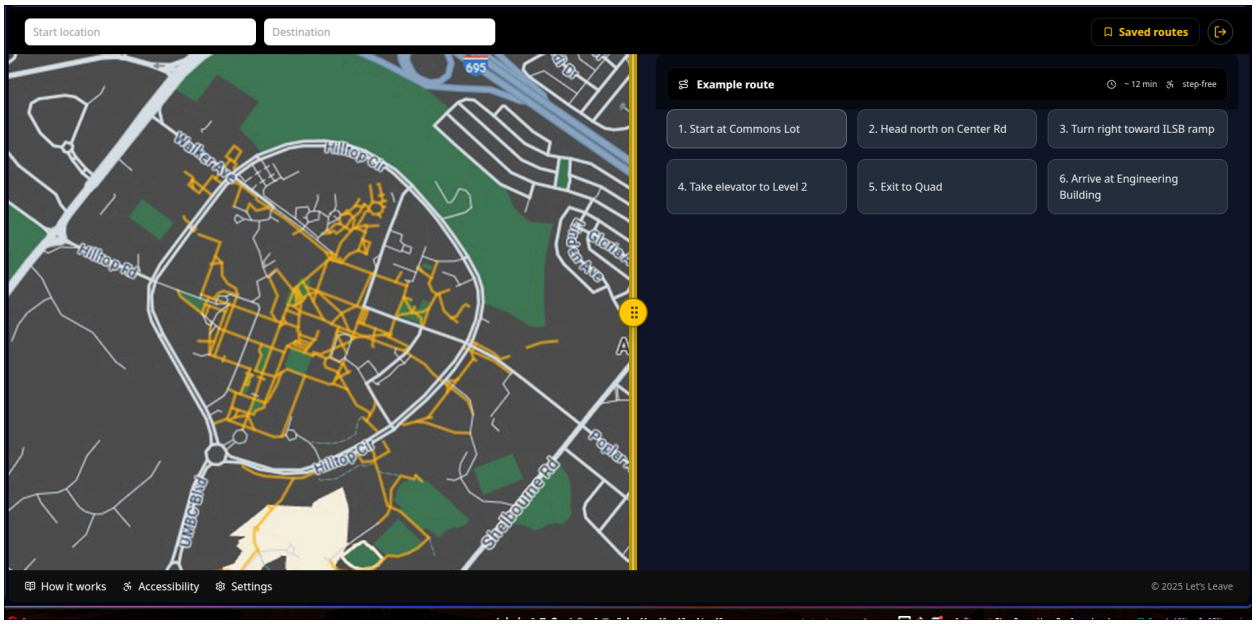



**Description:**
This screen allows users to authenticate using their UMBC Google credentials. The UI contains:

- Multiple accounts may be listed to choose from depending on how many accounts the user has linked (Only UMBC accounts will be authorized full privileges).
- Minimal text, per OAuth guidelines.

Upon successful login, the user is automatically redirected to the Main Page.

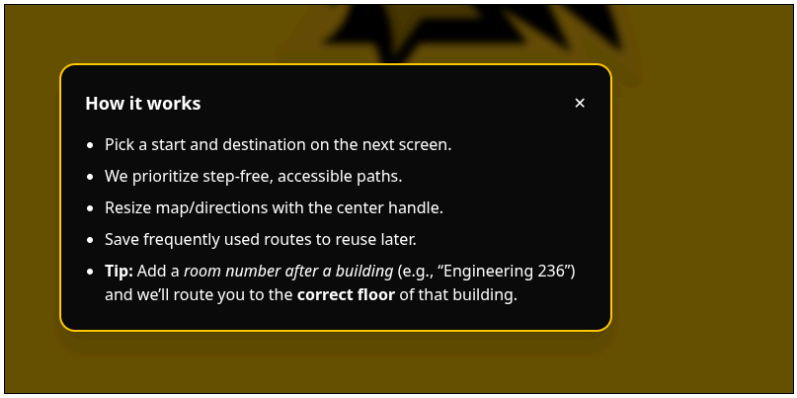
## Figure 3. Main Page (Map View + Instructions Panel)




Description:
The Main Page consists of two primary panels separated by a yellow grabber button:

1. Map Panel (Left Side)
    - Displays a Leaflet/OpenStreetMap interactive map.
    - Provides text fields for Start Location and Destination.
    - When both fields are valid, a route is rendered visually.
    - Zoom, pan, and waypoint markers appear directly on the map.
2. Instructions Panel (Right Side)
    - Lists turn by turn instructions for the selected route.
    - Scrollable column of step cards, each containing icons and short descriptions.
    - Automatically updates when the map route changes.
3. Center Split Panel Grabber
    - Allows users to widen or collapse either panel.
    - If dragged fully left or right, the hidden panel collapses into a tab for re-expansion.
4. Footer Navigation Bar
    - Persistent across the UI.
    - Links to informational pages without leaving the main session.

## Figure 4. How It Works Screen

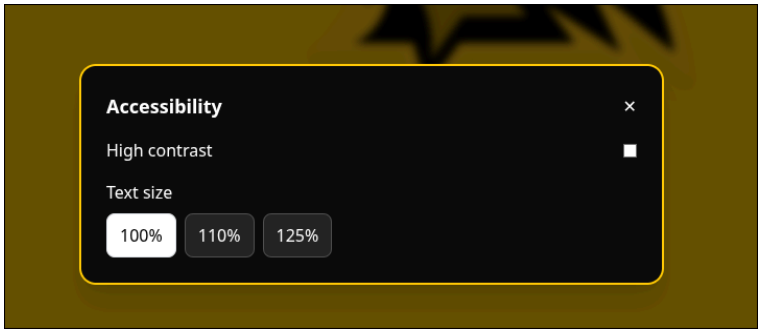



**Description:**
This informational screen explains how to use the routing system. It describes:

- How to input buildings and optional room numbers.
- How routes generate on the map.
- Optional helpful tips to make the user more efficient.

An "x" button returns to the Main Page.

## Figure 5. Accessibility Information Screen

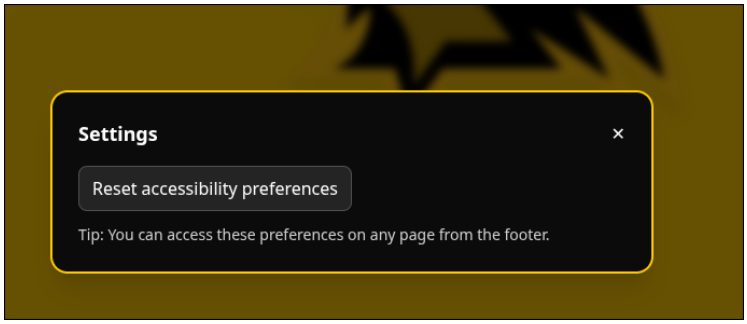



**Description:**
This page provides accessibility-related options:

- High contrast
- Text sizing

An "x" button returns to the Main Page.

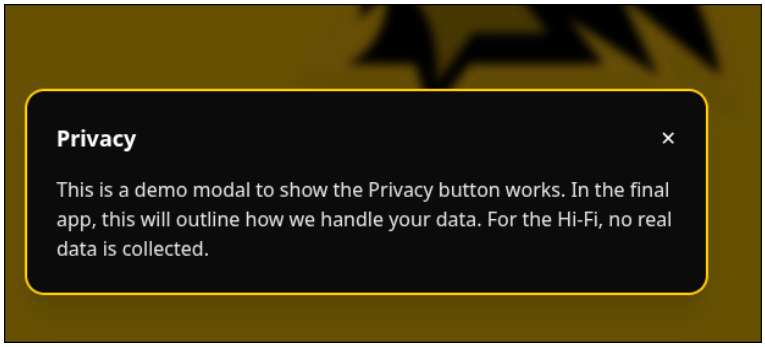
## Figure 6. Settings Screen




**Description:**
The Settings screen will allow customization of user preferences in the future but for now is extremely limited.

An “x” button returns to the Main Page.

### Figure 7. Privacy Screen




**Description:**
This page will display the system’s privacy policy. It is non-interactive except for scrolling. Includes an “x” button to return to the Main Page.

### Figure 8. Terms & Conditions Screen

**Description:**
The Terms & Conditions screen will contain the project’s usage guidelines. It follows the same standard informational-page layout and Includes an “x” button to return to the Main Page.

## 3.3 Summary

The navigation model of the UMBC Accessibility GPS is linear, consistent, and optimized for accessibility. All screens share unified UI components such as the footer bar, centered content areas, responsive layout scaling, and predictable “x” button behavior. Users can move easily between route generation, instructions, and informational pages without losing their current data or map state.

# 4. Data Validation

| **Location** | **Item** | **Data Type** | **Format** |
|---|---|---|---|
| Start Location Input Field | Input box for Route Start Building | String | {Building Name}{Room Number} |
| End Location Input Field | Input box for Route End Building | String | {Building Name}{Room Number} |

# Appendix A - Agreement Between Customer and Contractor

By signing this document, both the customer and the developers agree to the details specified in this document. Specifically, the scope of the project, the functional requirements listed, and the non-functional requirements listed. By signing this document the customer also agrees to cooperate with the development team for any future questions they may have.

If the customer wishes to make any changes to this document, they must first talk to the development team to make those changes. Then, both parties must agree to those changes and re-sign this document. Then the version number will be updated along with all of the changes documented.

<u>Customer</u>

| Name | Signature | Date | Comments |
|---|---|---|---|
| | | | |

<u>Team</u>

| Name | Signature | Date | Comments |
|---|---|---|---|
| Aidan Denham | *Aidan Denham* | 11/22/2025 | |
| Celestine Sumah | *Celestine Sumah* | 11/22/2025 | n/a |

| | | | |
|---|---|---|---|
| Ethan Michalik | Ethan Michalik | 11/22/2025 | |
| Alex Marbut | Alex Marbut | 11/22/2025 | |

# Appendix B - Peer Review Sign-off

By signing this document, all members of the development team have reviewed this document and agreed to its contents and format. If there are any major points of contention, please refrain from signing until they have been addressed.

| Name | Signature | Date | Comments |
|---|---|---|---|
| Celestine Sumah | Celestine Sumah | 11/22/2025 | n/a |
| Aidan Denham | Aidan Denham | 11/22/2025 | |
| Ethan Michalik | Ethan Michalik | 11/22/2025 | |
| Alex Marbut | Alex Marbut | 11/22/2025 | |

# Appendix C - Document Contributions

Contributions by: Aidan Denham, Alex Marbut, Celestine Sumah, Ethan Michalik